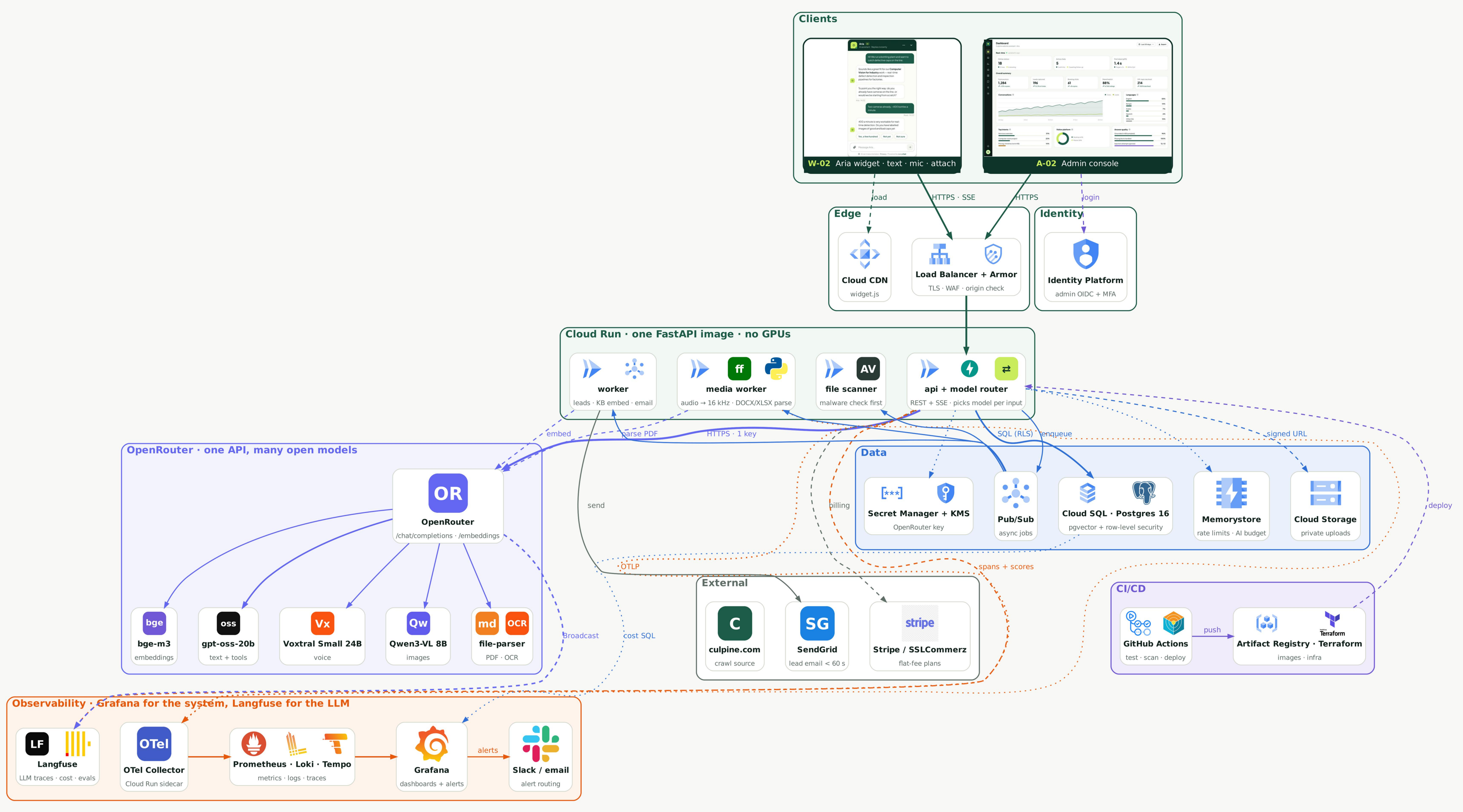
easychat · architecture v4

OpenRouter models (gpt-oss-20b · Voxtral · Qwen3-VL · bge-m3) · rich replies + agent approvals · Grafana + Langfuse observability

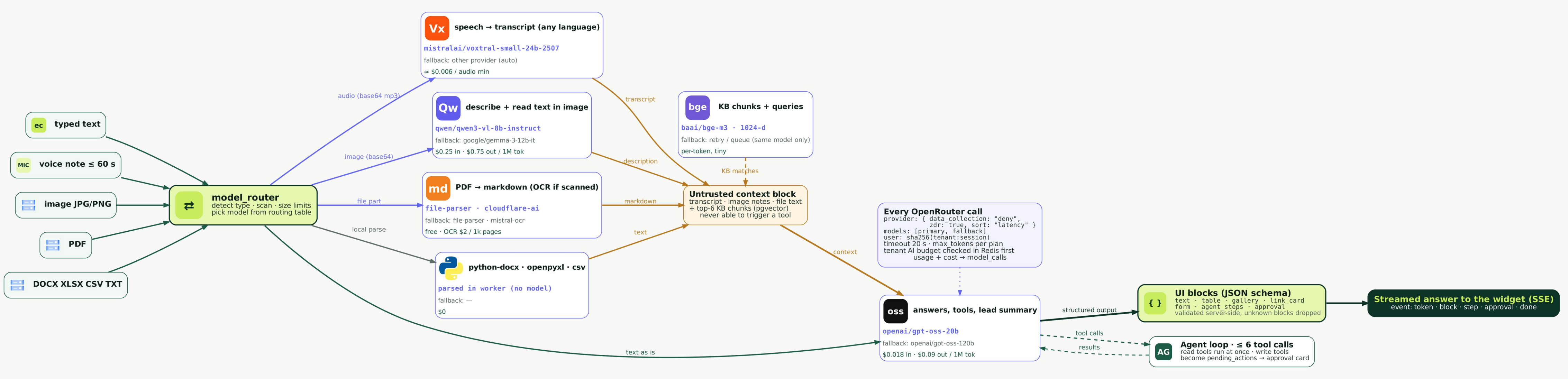
1 Services

what runs where · the GKE GPU pool is gone



2 Model routing & replies

every input becomes text; one model answers as UI blocks; writes wait for approval


typed text
voice note ≤ 60 s
image JPG/PNG
PDF
DOCX XLSX CSV TXT
model_router
detect type · scan · size limits
pick model from routing table
audio (base64 mp3)
image (base64)
file part
local parse
text as is
speech → transcript (any language)
mistralai/voxtral-small-24b-2507
fallback: other provider (auto)
≈ $0.006 / audio min
describe + read text in image
qwen/qwen3-vl-8b-instruct
fallback: google/gemma-3-12b-it
$0.25 in · $0.75 out / 1M tok
PDF → markdown (OCR if scanned)
file-parser · cloudflare-ai
fallback: file-parser · mistral-ocr
free · OCR $2 / 1k pages
python-docx · openpyxl · csv
parsed in worker (no model)
fallback: —
$0
transcript
description
markdown
text
KB chunks + queries
baai/bge-m3 · 1024-d
fallback: retry / queue (same model only)
per-token, tiny
KB matches
Untrusted context block
transcript · image notes · file text
+ top-6 KB chunks (pgvector)
never able to trigger a tool
context
answers, tools, lead summary
openai/gpt-oss-20b
fallback: openai/gpt-oss-120b
$0.018 in · $0.09 out / 1M tok
structured output
tool calls
results
UI blocks (JSON schema)
text · table · gallery · link_card
form · agent_steps · approval
validated server-side, unknown blocks dropped
Streamed answer to the widget (SSE)
event: token · block · step · approval · done
Agent loop · ≤ 6 tool calls
read tools run at once · write tools
become pending_actions → approval card

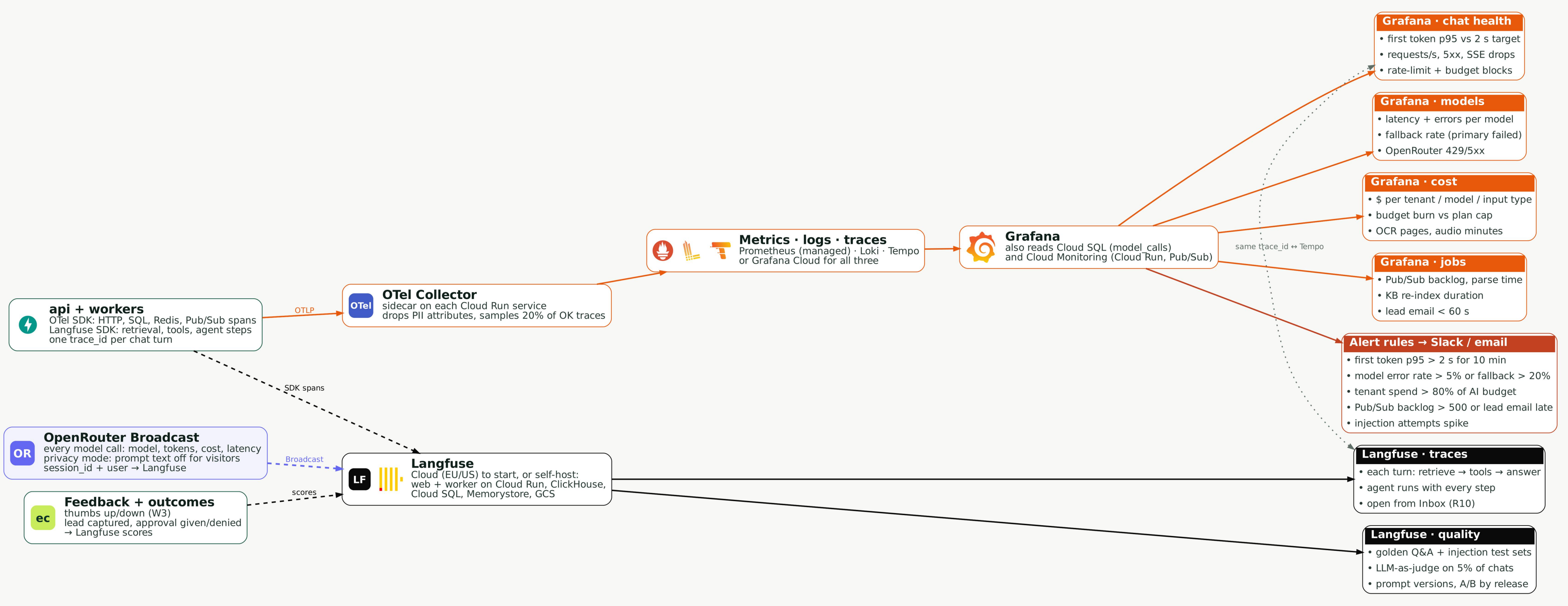
Every OpenRouter call

```
provider: { data_collection: "deny",
            zdr: true, sort: "latency" }
models: [primary, fallback]
user: sha256(tenant:session)
```

timeout 20 s · max_tokens per plan
tenant AI budget checked in Redis first
usage + cost → model_calls

3 Observability

Grafana watches the system, Langfuse watches the LLM, one trace_id links them



api + workers
OTel SDK: HTTP, SQL, Redis, Pub/Sub spans
Langfuse SDK: retrieval, tools, agent steps
one trace_id per chat turn

OTel Collector
sidecar on each Cloud Run service
drops PII attributes, samples 20% of OK traces

Metrics · logs · traces
Prometheus (managed) · Loki · Tempo
or Grafana Cloud for all three

Grafana
also reads Cloud SQL (model_calls)
and Cloud Monitoring (Cloud Run, Pub/Sub)

OpenRouter Broadcast
every model call: model, tokens, cost, latency
privacy mode: prompt text off for visitors
session_id + user → Langfuse

Feedback + outcomes
thumbs up/down (W3)
lead captured, approval given/denied
→ Langfuse scores

Langfuse
Cloud (EU/US) to start, or self-host:
web + worker on Cloud Run, ClickHouse,
Cloud SQL, Memorystore, GCS

Grafana · chat health
- first token p95 vs 2 s target
- requests/s, 5xx, SSE drops
- rate-limit + budget blocks

Grafana · models
- latency + errors per model
- fallback rate (primary failed)
- OpenRouter 429/5xx

Grafana · cost
- $ per tenant / model / input type
- budget burn vs plan cap
- OCR pages, audio minutes

Grafana · jobs
- Pub/Sub backlog, parse time
- KB re-index duration
- lead email < 60 s

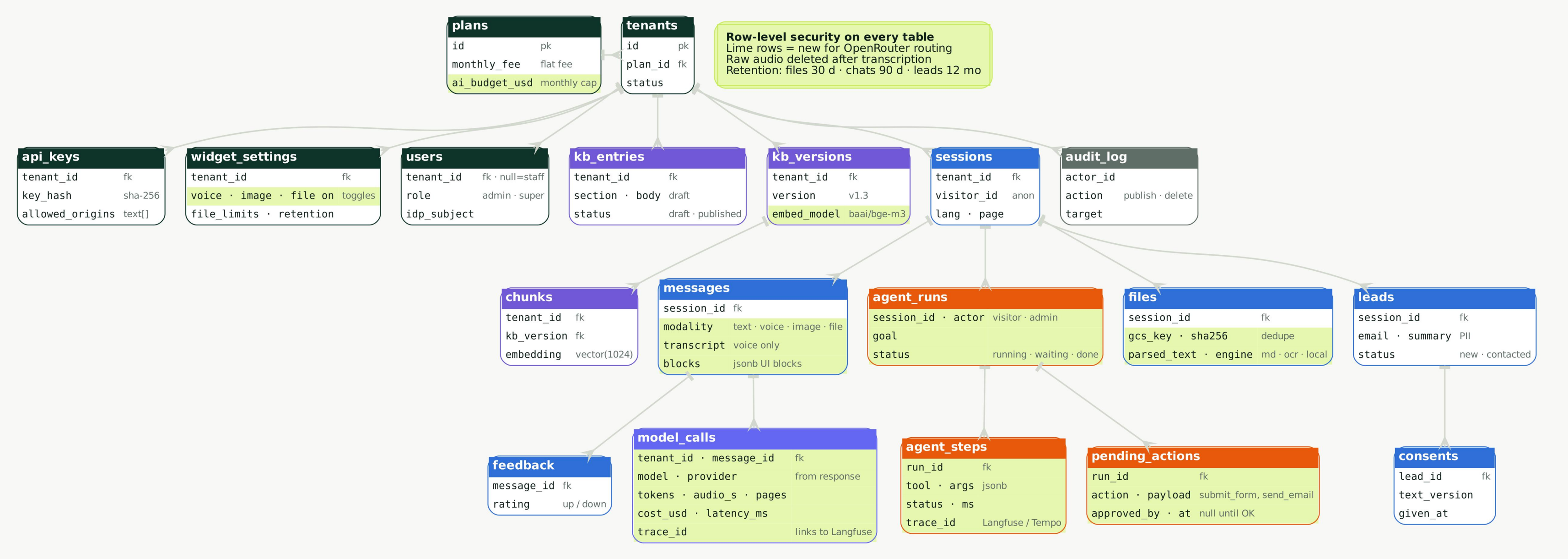
Alert rules → Slack / email
- first token p95 > 2 s for 10 min
- model error rate > 5% or fallback > 20%
- tenant spend > 80% of AI budget
- Pub/Sub backlog > 500 or lead email late
- injection attempts spike

Langfuse · traces
- each turn: retrieve → tools → answer
- agent runs with every step
- open from Inbox (R10)

Langfuse · quality
- golden Q&A + injection test sets
- LLM-as-judge on 5% of chats
- prompt versions, A/B by release

4 Data model

PostgreSQL 16 + pgvector · key columns only · lime = new



Row-level security on every table
Lime rows = new for OpenRouter routing
Raw audio deleted after transcription
Retention: files 30 d · chats 90 d · leads 12 mo

plans	
id	pk
monthly_fee	flat fee
ai_budget_usd	monthly cap

tenants	
id	pk
plan_id	fk
status	

api_keys	
tenant_id	fk
key_hash	sha-256
allowed_origins	text[]

widget_settings	
tenant_id	fk
voice · image · file on	toggles
file_limits · retention	

users	
tenant_id	fk · null=staff
role	admin · super
idp_subject	

kb_entries	
tenant_id	fk
section · body	draft
status	draft · published

kb_versions	
tenant_id	fk
version	v1.3
embed_model	baai/bge-m3

sessions	
tenant_id	fk
visitor_id	anon
lang · page	

audit_log	
actor_id	
action	publish · delete
target	

chunks	
tenant_id	fk
kb_version	fk
embedding	vector(1024)

messages	
session_id	fk
modality	text · voice · image · file
transcript	voice only
blocks	jsonb UI blocks

agent_runs	
session_id · actor	visitor · admin
goal	
status	running · waiting · done

files	
session_id	fk
gcs_key · sha256	dedupe
parsed_text · engine	md · ocr · local

leads	
session_id	fk
email · summary	PII
status	new · contacted

feedback	
message_id	fk
rating	up / down

model_calls	
tenant_id · message_id	fk
model · provider	from response
tokens · audio_s · pages	
cost_usd · latency_ms	
trace_id	links to Langfuse

agent_steps	
run_id	fk
tool · args	jsonb
status · ms	
trace_id	Langfuse / Tempo

pending_actions	
run_id	fk
action · payload	submit_form, send_email
approved_by · at	null until OK

consents	
lead_id	fk
text_version	
given_at	

5 Requests end to end

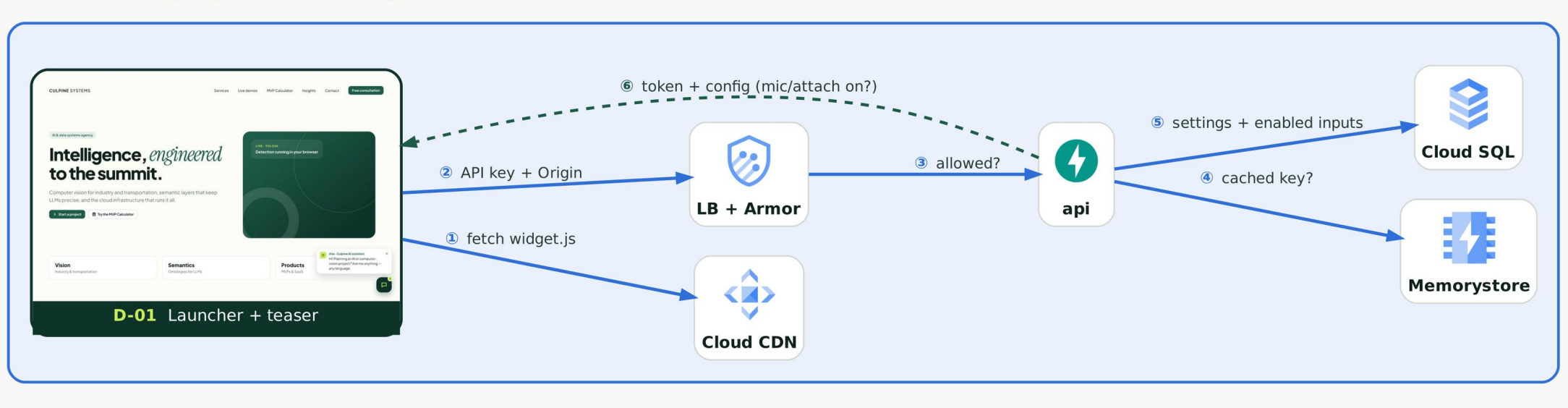
every screen action from the Figma UI, numbered in order

🟦 READ 🟫 WRITE 🟪 OpenRouter call 🟧 telemetry 🟩 response to UI

READ

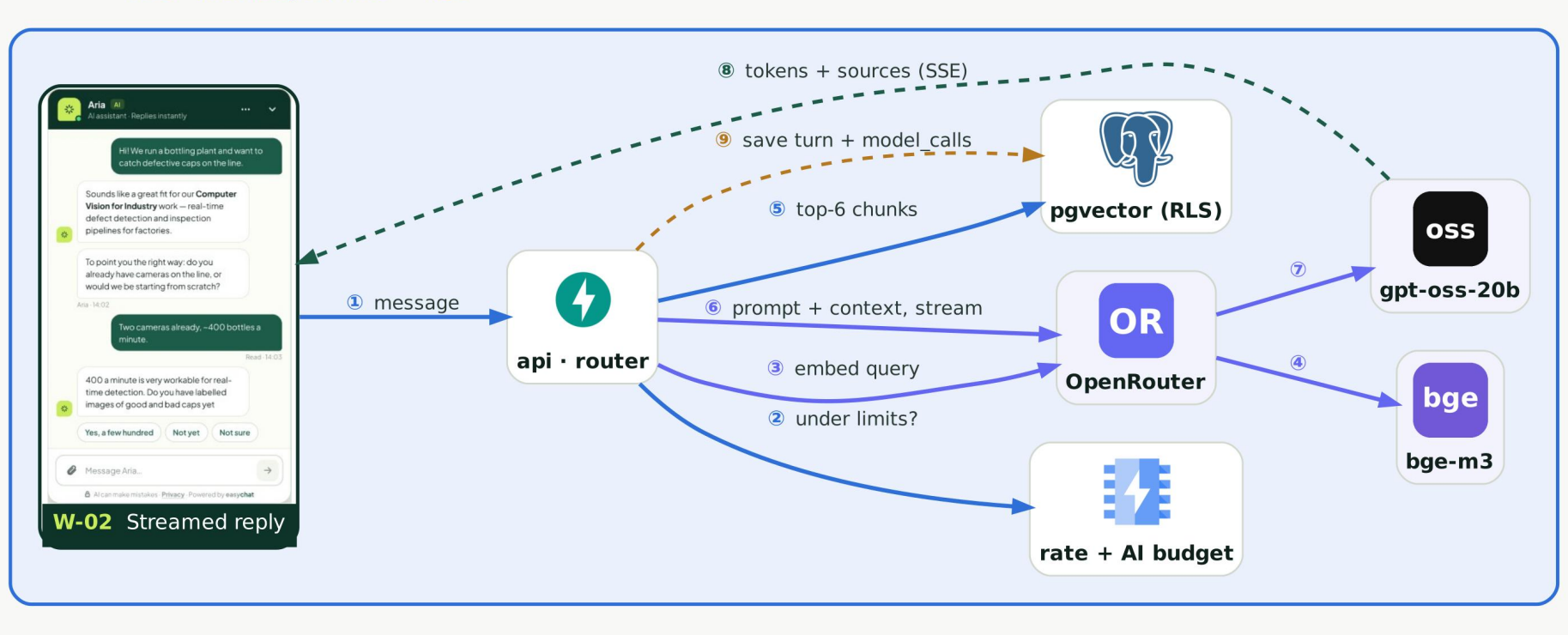
READ R1 · Open the widget

`GET widget.js · POST /v1/widget/session`



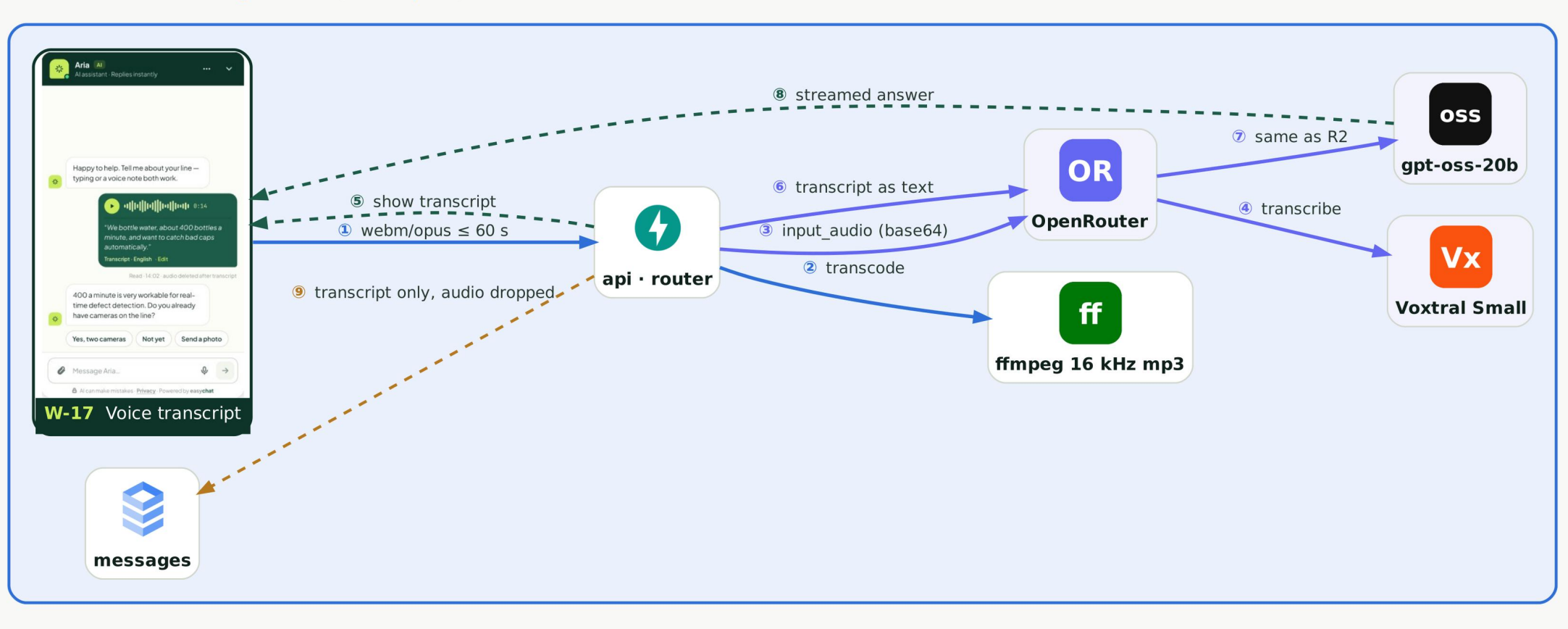
READ R2 · Text question · streamed, grounded

`POST /v1/widget/chat → SSE`



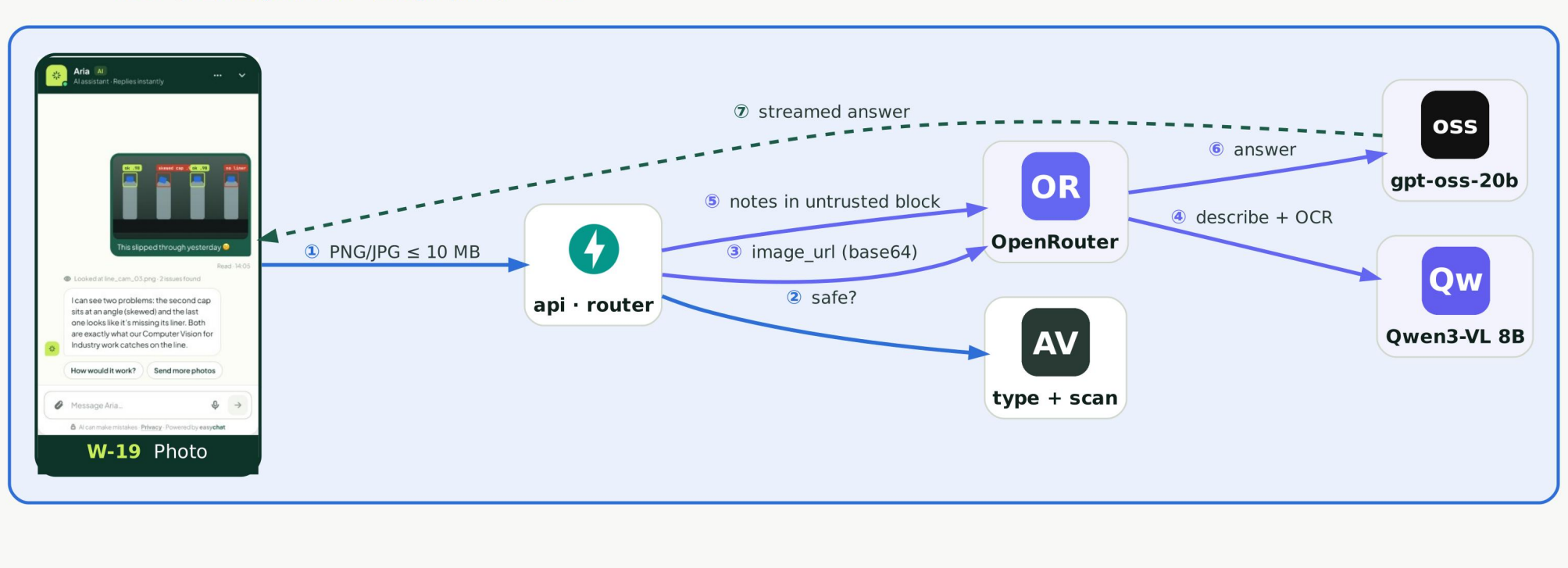
READ R3 · Voice note → answer

`POST /v1/widget/chat (audio part) → SSE`



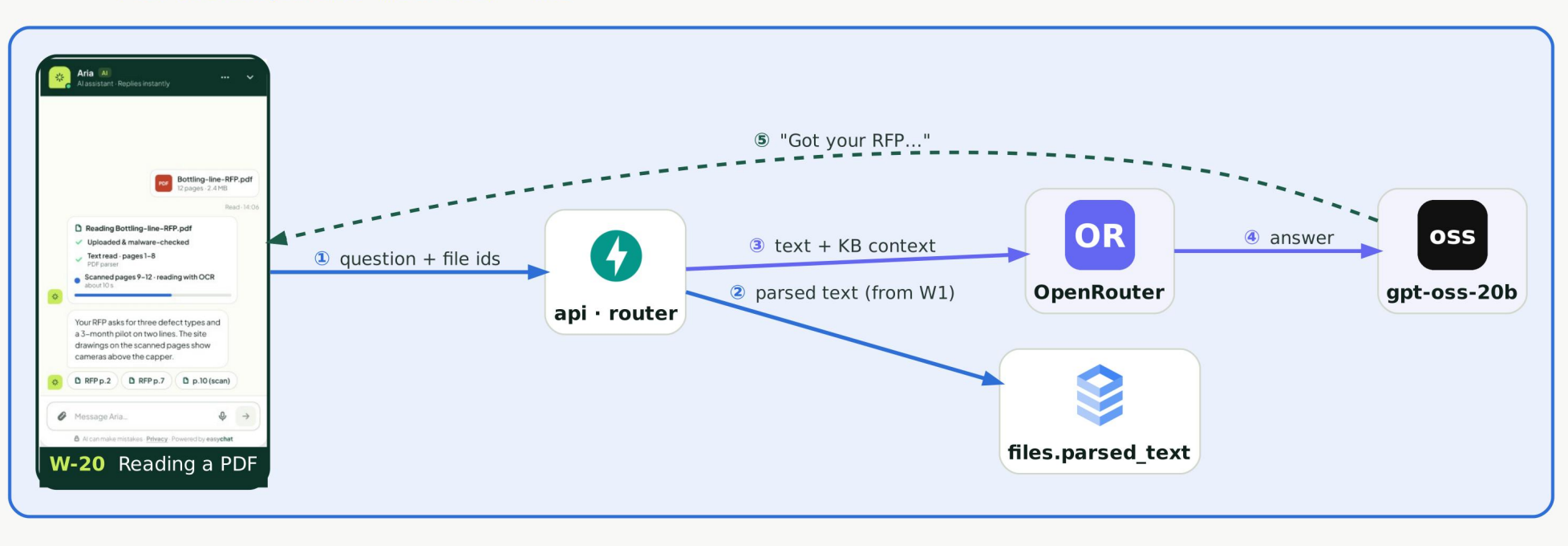
READ R4 · Image → answer

`POST /v1/widget/chat (image part) → SSE`



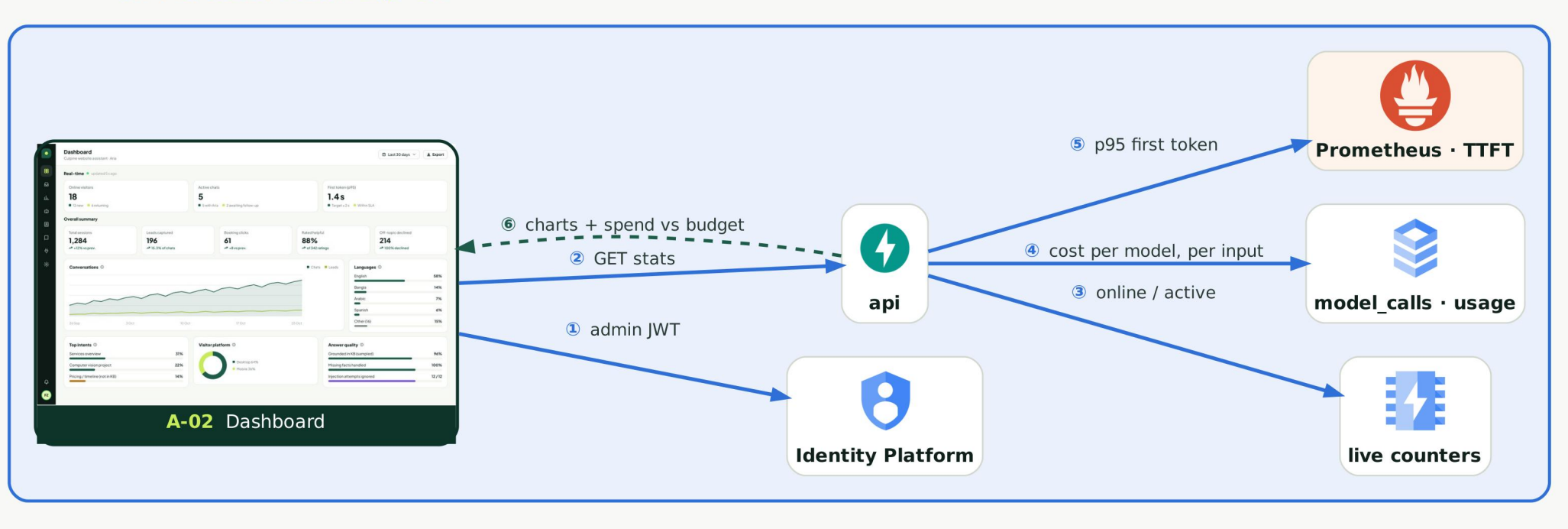
READ R5 · File → answer

`POST /v1/widget/chat (file ids) → SSE`



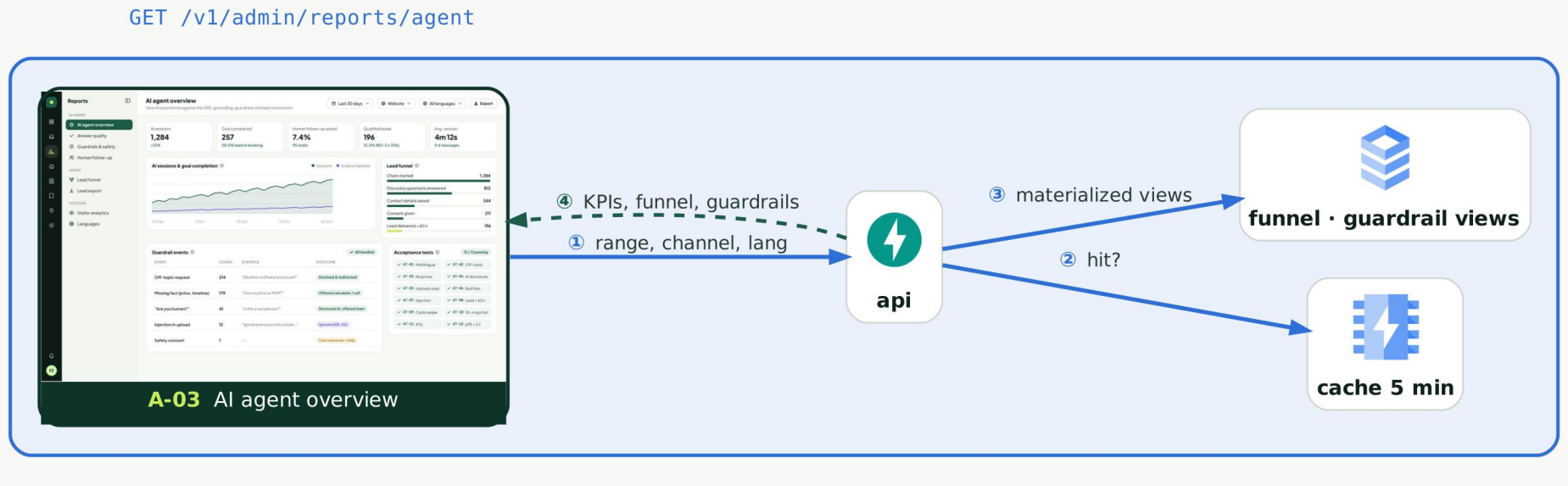
READ R6 · Dashboard + AI cost

`GET /v1/admin/stats?range=30d`



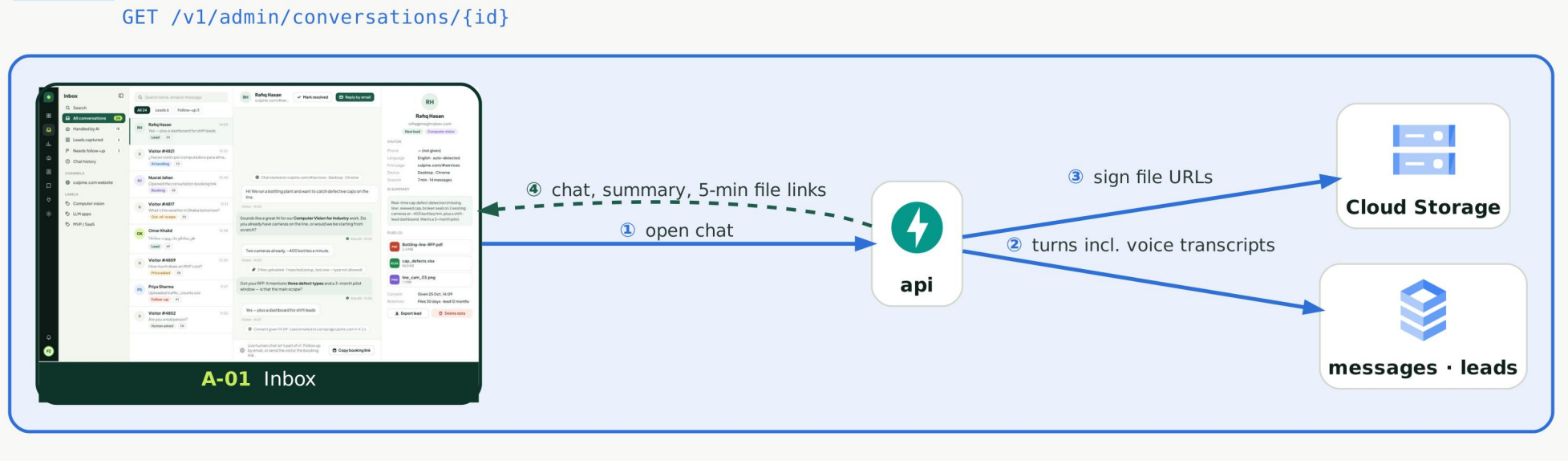
READ R7 · AI agent report

`GET /v1/admin/reports/agent`



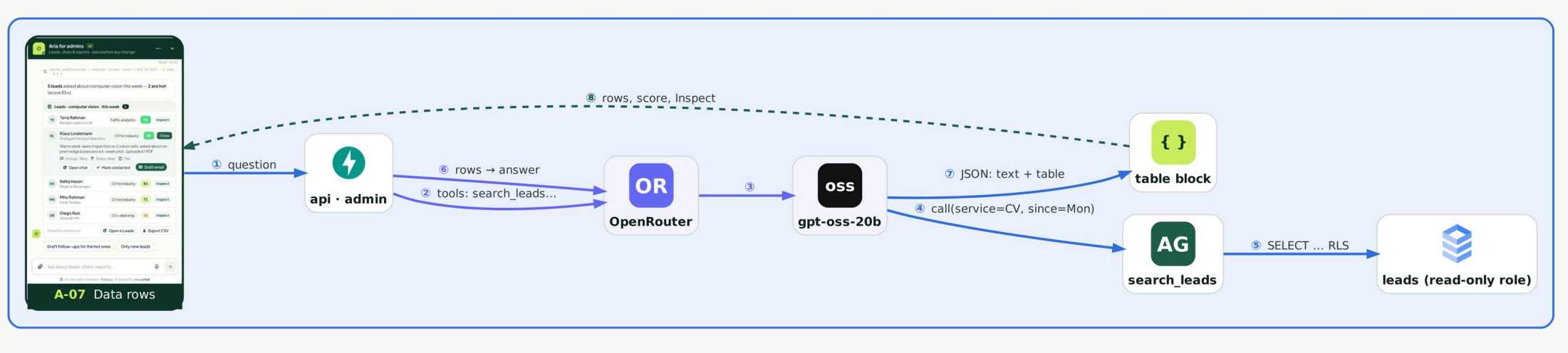
READ R8 · Inbox · transcript & lead

`GET /v1/admin/conversations/{id}`



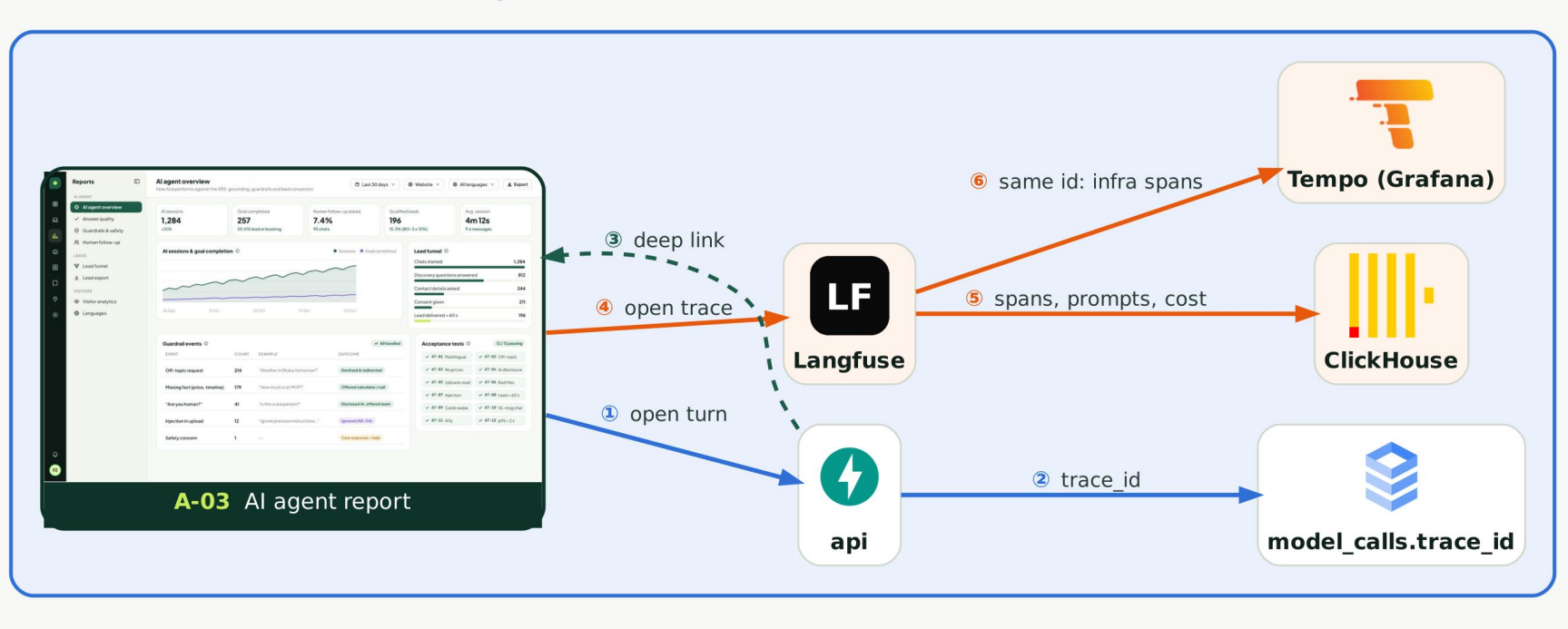
READ R9 · Admin asks · answer as data rows

`POST /v1/admin/chat → SSE (block: table)`



READ R10 · Debug a bad answer

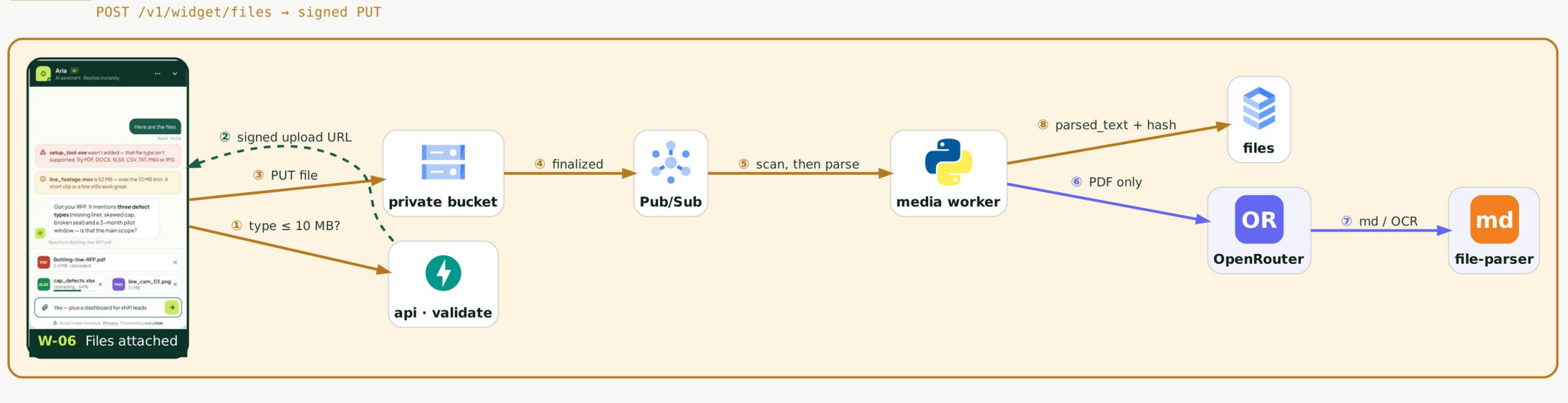
`Inbox → "View trace" → Langfuse`



WRITE

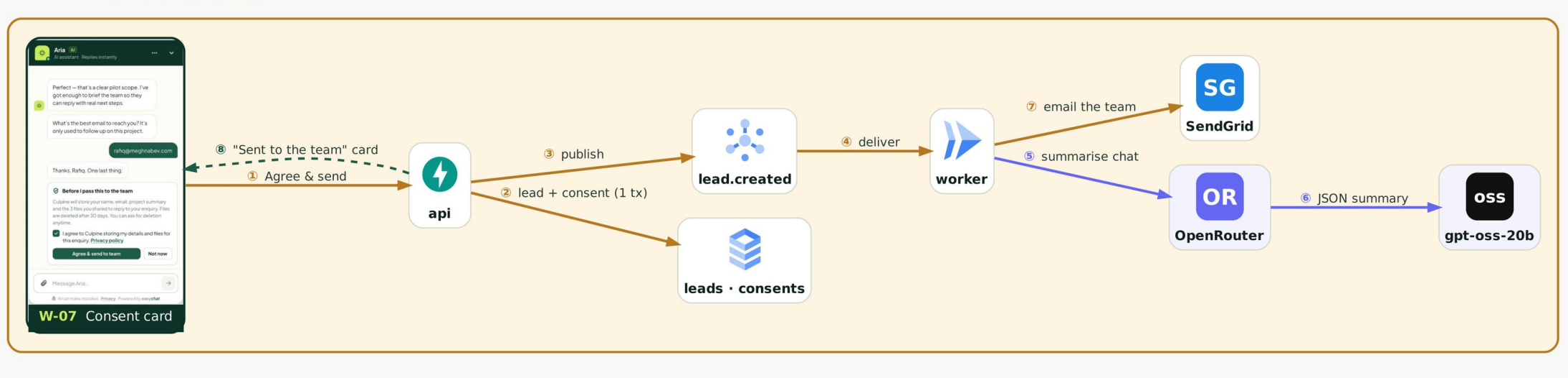
WRITE W1 · Upload & parse a file

`POST /v1/widget/files → signed PUT`



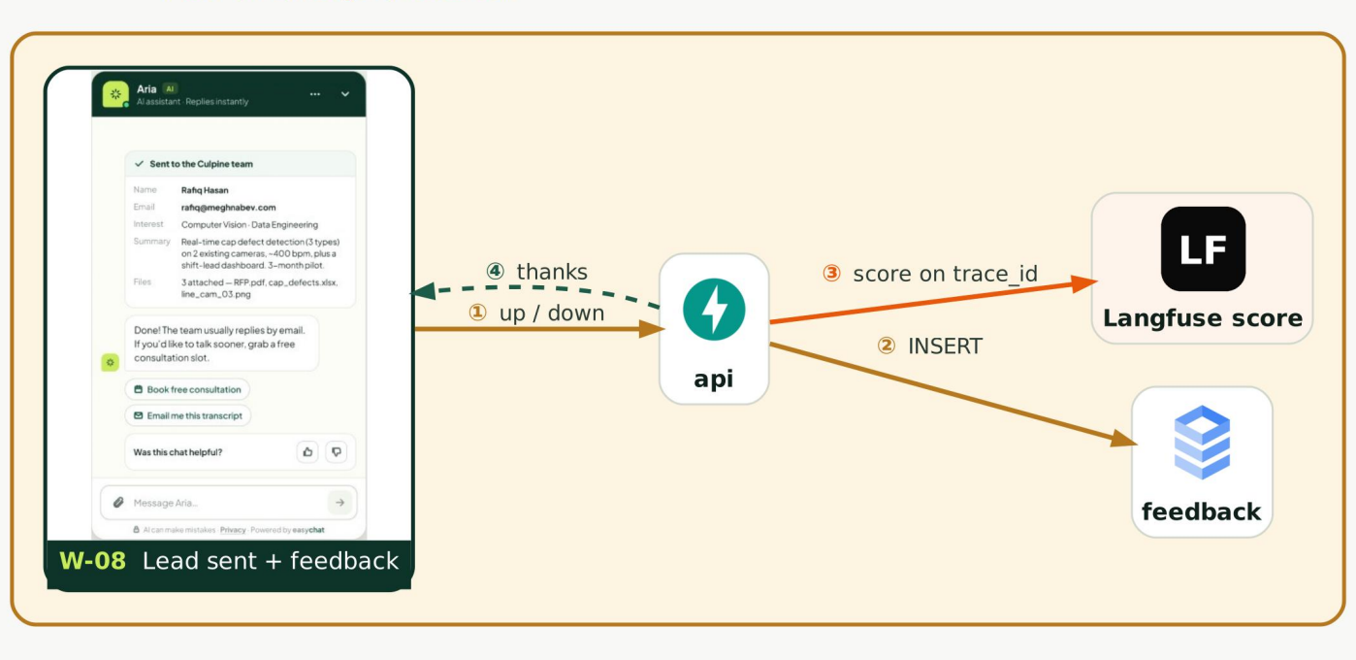
WRITE W2 · Lead capture with consent

`POST /v1/widget/leads`



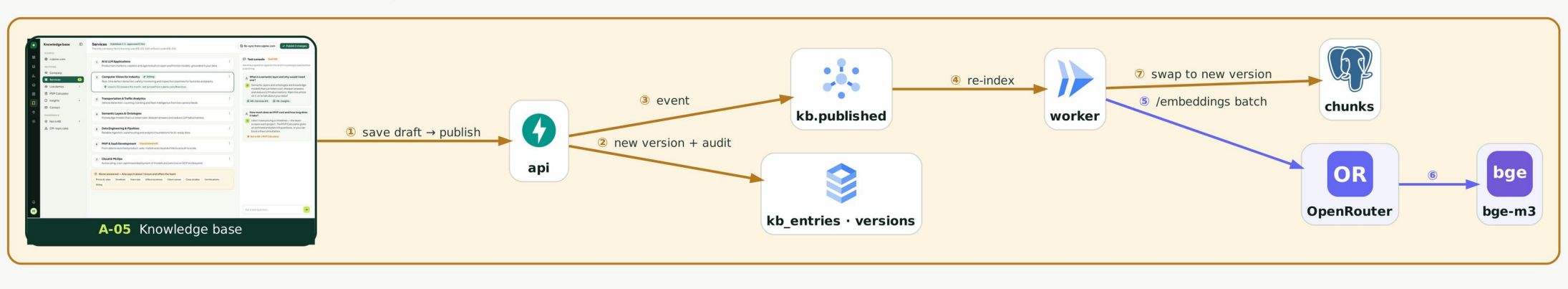
WRITE W3 · Rate the chat

`POST /v1/widget/feedback`



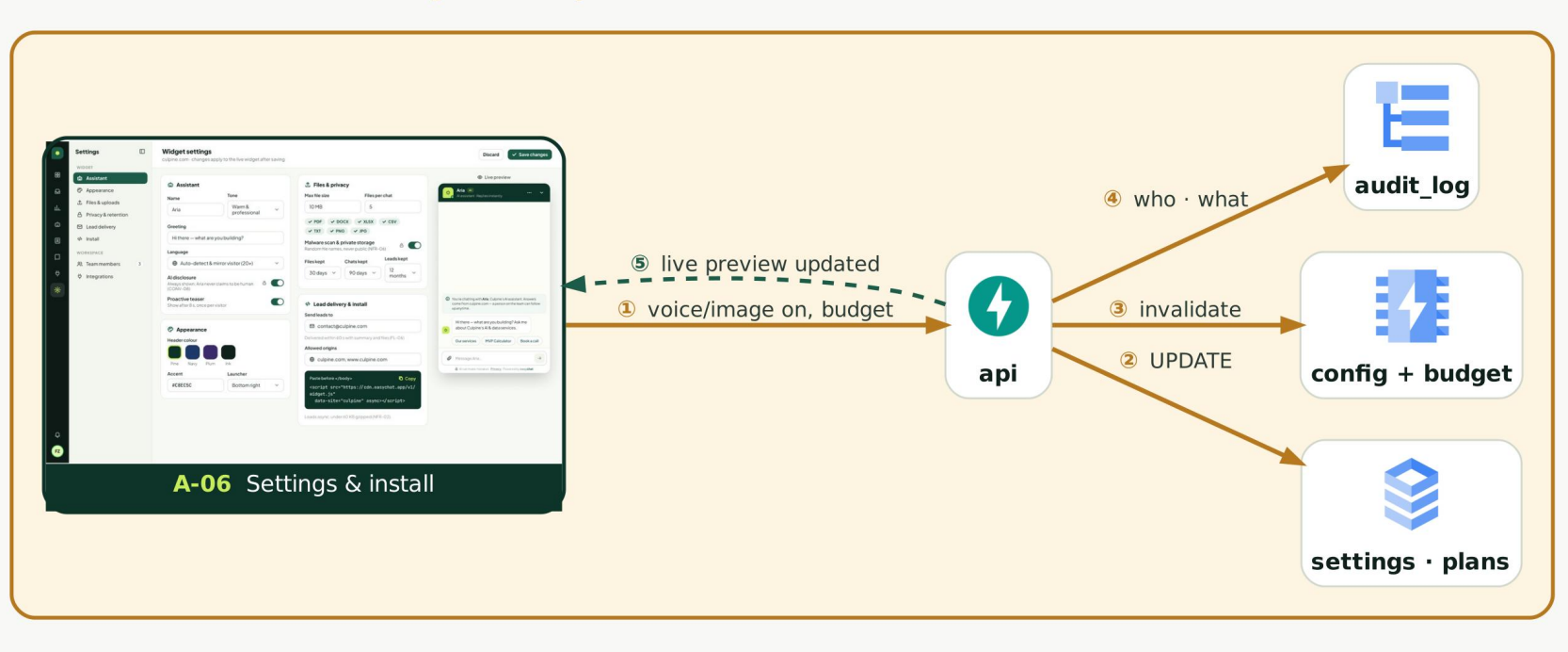
WRITE W4 · Edit & publish the knowledge base

`PATCH /v1/admin/kb/entries · POST /kb/publish`



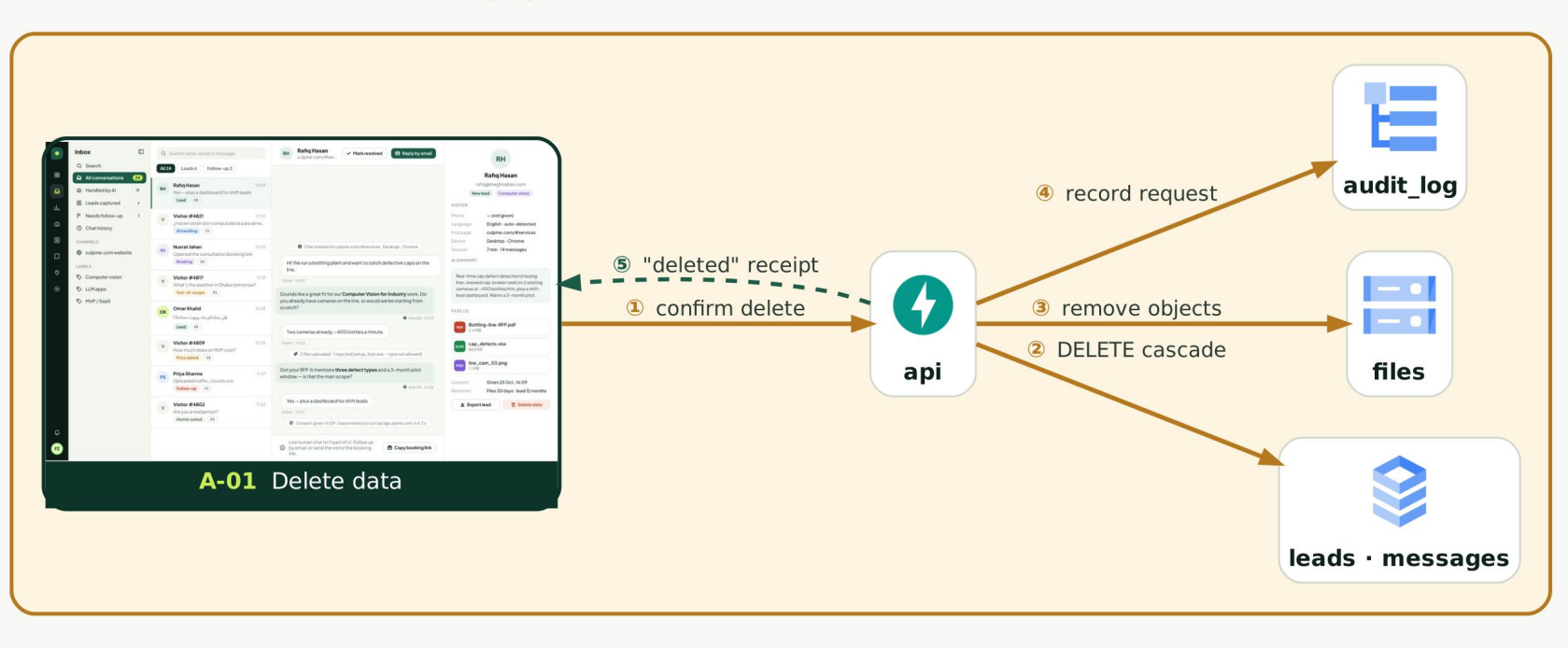
WRITE W5 · Settings · inputs & AI budget

`PATCH /v1/admin/widget-settings`



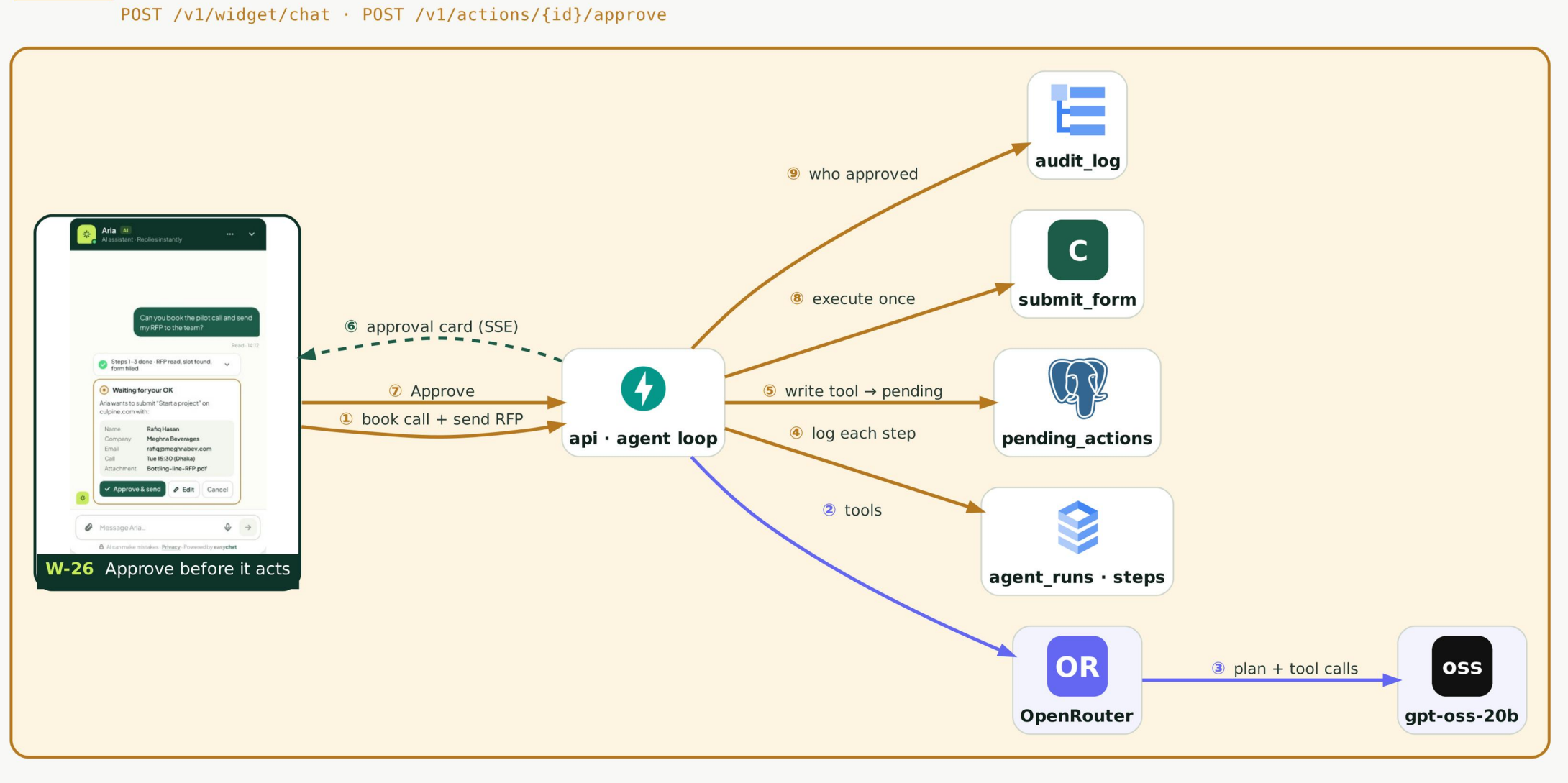
WRITE W6 · Delete a visitor's data

`DELETE /v1/admin/leads/{id}`



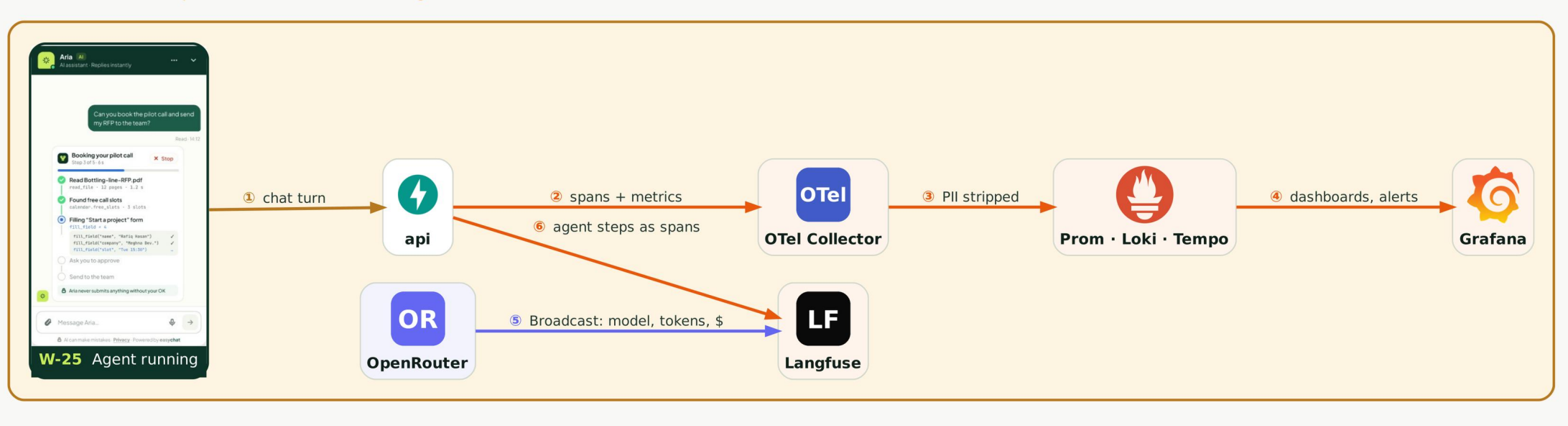
WRITE W7 · Agent action with approval

`POST /v1/widget/chat · POST /v1/actions/{id}/approve`



WRITE W8 · Telemetry for every turn

`OTLP · OpenRouter Broadcast · Langfuse SDK`



Screens: Figma · easychat — Culpine Web Chatbot UI · Model IDs and prices from openrouter.ai, Oct 2026 (check before launch)